

No detectable hepatocyte de-zonation across acquisition-matched biopsy MASLD: tissue-source confounding in single-nucleus data

Brief re-analysis report. Built entirely from the fresh raw-count re-analysis (FINDINGS.md); independent of all earlier project reports. Dataset: Gribben et al., Nature 2024 (GSE202379).

Abstract

Single-nucleus RNA-seq of human liver across metabolic dysfunction–associated steatotic liver disease (MASLD) has been interpreted as showing that hepatocytes **lose their zonation** during disease, with end-stage hepatocytes co-expressing pericentral and periportal markers. Re-analysing the same cohort (47 donors, 69,426 hepatocyte nuclei) on **raw RNA UMI counts** with **donor-level inference**, we find that **disease stage is perfectly collinear with tissue source**: every end-stage sample is a transplant explant, while all milder disease samples are needle biopsies. Explant tissue carries an organ-wide procurement-stress signature (immediate-early and heat-shock genes elevated 5–21× across *every* cell lineage, including endothelium), so end-stage cannot be read as a disease endpoint. Restricting to acquisition-matched needle-biopsy donors and testing every **marker-count mechanism detectable in spatially-blind snRNA** by which zonation could collapse — pericentral-cell depletion, within-cell program dimming, marker co-expression, gradient compression, identity turn-off, and compositional shift — we detect **no change across fibrosis Fo–F4 or NAS activity**, robust to depth-matching budget, Monte-Carlo draw, anchor definition, and thresholds. The design is powered to exclude near-total loss but not finer change. The study **excludes near-total marker-count collapse but is underpowered for moderate, focal, or spatially restricted de-zonation** — "no detectable change" is not "quantified preservation." We conclude that the single-nucleus/transcriptional evidence for progressive hepatocyte de-zonation in MASLD is confounded by tissue procurement; the dramatic end-stage signal is confined to heterogeneous transplant explants and is inseparable from organ-wide procurement stress, so this cohort **cannot identify it as progressive disease** (it neither proves nor disproves disease biology). The protein-imaging and organoid evidence is not addressed by these data. Transcriptome-wide donor-level differential expression confirms the hepatocyte zonation and xenobiotic programs are unchanged across biopsy fibrosis (every zonation/detox gene FDR > 0.79); the 23 genes that do trend are dominated by non-parenchymal (stromal/immune) ambient signal, the only clearly hepatocyte-intrinsic change being a modest acute-phase response (*A2M*) — not de-zonation.

The research question, and the answer

This project was framed precisely: *using the healthy porto-central zonation ruler from Paper 2, can we assign a quantitative pseudo-zonation coordinate to every spatially-blind Paper-1 hepatocyte, and so measure how — and how much — hepatocyte zonation degrades as MASLD advances from healthy to end-stage, and is that degradation linked to the transdifferentiation Paper 1 reported?*

In acquisition-matched tissue, the answer is **no** for the substantive clauses (and *unanswerable* — not merely negative — for the transdifferentiation link):

1. **A coordinate can be assigned — but it is the wrong instrument.** A Paper-2-derived signature does place each hepatocyte on a pseudo-zonation axis; however, that fitted coordinate, and the marker-correlation statistics built on it, are precisely what convert differences in sequencing depth, cell number and tissue state into an *apparent* collapse. We therefore measure zonation directly, in raw molecule counts per nucleus, rather than through a fitted ruler.
2. **No measurable degradation across disease.** Across acquisition-matched biopsy MASLD — healthy/atlas and end-stage explants excluded as non-comparable — hepatocyte zonation does not measurably degrade from F0 to cirrhotic F4 by any count-based mechanism. The apparent "healthy → end-stage" degradation collapses onto the biopsy-versus-explant tissue axis, not the disease axis.
3. **No establishable link to transdifferentiation.** Within hepatocyte nuclei, biliary/transdifferentiation markers (*KRT7*, *KRT19*, *EPCAM*, *SOX4*) remain at background (~0.4–1.3% detection) and flat across biopsy F1–F4, rising only in end-stage explants — the *same* confounded group in which the apparent de-zonation appears. Both signatures co-localize to the explant artifact; in comparable tissue neither is detectable, so a zonation-degradation ↔ transdifferentiation link cannot be established here (nor is it refuted).

Overall: the dataset does not support measurable, disease-driven hepatocyte de-zonation in MASLD, nor a link to the reported transdifferentiation; the apparent signal is a tissue-procurement confound. (We address the single-nucleus/transcriptional evidence; we do not refute the orthogonal protein-imaging and organoid evidence — see Discussion.)

Data provenance and point of departure

Paper 1 (Gribben et al., GSE202379) — what was deposited, and what we took. The authors' pipeline (their Methods) mapped reads with CellRanger v5.0.0 (GRCh38.p13, `--include-introns`), applied Seurat v4.0.3 quality control (≥ 200 features; exclude nuclei with $<1,000$ / <800 features or $>10\%$ mitochondrial/ribosomal counts; remove $>8,000$ -feature doublets; and **remove mitochondrial and ribosomal protein-coding genes**), then normalized with **SCTransform**, ran PCA + Harmony integration + Louvain clustering, and annotated cell types by marker genes. The deposited Seurat object therefore carries **two assays**: a **raw RNA assay** (31,257 genes — `counts` = integer molecules, `data` = log-normalized) and a **processed SCT assay** (30,117 genes — `counts` / `data` / `scale.data`, the SCTransform output), with **SCT set as the default assay**, plus 52 metadata columns and the authors' SCT/Harmony cluster solutions.

For molecule-level inference we took **only the raw RNA assay counts layer**, verified raw by nine committed sanity checks (`prep/06_sanity_raw.py`): integer-valued; per-nucleus library size 920–49,854 (vs the depth-equalized SCT band 2,908–6,292); value-level different from the SCT matrix; ALB-dominant; $n = 69,426$ hepatocytes; and no silent gene drop. We retained a 45-gene panel (one declared gene, the ribosomal *RPLP0*, is legitimately absent — the authors removed ribosomal genes at QC), each nucleus's full raw library size, and the donor/stage/lobe/annotation and clinical/staging metadata (re-joined from the metadata table). We **dropped** the SCT assay, the Harmony embeddings, and the authors' clusters, and re-derived all normalization ourselves

(transparent depth-matching). This is our point of departure: we take the authors' *raw* layer and nothing downstream of their normalization.

The first (failed) analysis used the wrong layer — a post-mortem. An earlier extraction pulled `GetAssayData(default = SCT, "counts")` — the authors' **SCTransform-corrected pseudo-counts** (depth-equalized model output, *not* observed molecules) — and scored a *relative* zonation coordinate on them ($z(PC) - z(PP)$), even "controlling for depth" by binomially thinning the already-depth-corrected SCT counts (circular: their depth had been regressed out). SCTransform is a legitimate normalization and a defensible input for a signature *score*, so this was not simply "wrong data." The failures were two, on different layers: **(i)** the relative z-scored-coordinate / marker-correlation **approach** itself, which by construction erases absolute expression level (it cannot see a program switch off) and absorbs sequencing depth, cell number, and **tissue source** into an apparent "collapse" — a confound present on *any* substrate; and **(ii)** remaining on the SCT (corrected) matrix once the question became molecule-counting, where corrected pseudo-counts are the wrong substrate. The present analysis fixes both: direct count-based tests on raw molecules, with our own transparent depth control.

Paper 2 (healthy zonation reference). Paper 2 framed the question (a healthy porto-central "ruler") but is **not a data dependency of this result**: the pericentral ↔ periportal axis here is defined by **canonical literature markers** (*GLUL/CYP3A4* vs *ASS1/CPS1/PCK1/HAL/ALDOB*), not a Paper-2-fit signature or fitted pseudo-coordinate. (The abandoned ruler approach did depend on Paper-2-derived signatures; we do not use them.)

Introduction

The hepatic lobule is spatially **zonated**: pericentral (zone 3) hepatocytes around the central vein run a Wnt/β-catenin-driven program (glutamine synthetase *GLUL*, drug-metabolizing cytochromes), while periportal (zone 1) hepatocytes around the portal triad run urea-cycle and gluconeogenic programs (*ASS1*, *CPS1*, *PCK1*). Whether this organization is lost in chronic liver disease is biologically important and clinically suggestive.

A widely-cited single-nucleus study of MASLD progression reported that "hepatocytes lose their zonation," with end-stage hepatocytes co-expressing pericentral and periportal markers. The transcriptional evidence was a **breakdown in the correlation between pericentral and periportal markers** across disease stage. Marker correlation is a relative summary statistic: it compresses a joint distribution into one number and is sensitive to cell number, sequencing depth, and tissue state, none of which it reports. We asked whether the de-zonation signal survives a strictly count-based, donor-level, confound-controlled re-analysis of the same data.

Results

1. Disease stage is confounded with tissue source; end-stage is not a usable disease endpoint

The processed expression matrix distributed with the dataset is SCTransform-corrected (per-nucleus totals equal `nCount_SCT` exactly, compressed to ~3,000–5,700), not raw molecules. We therefore

extracted **raw RNA-assay UMI counts** from the original Seurat object for a curated panel of zonation, stress, identity and control genes (library sizes 920–49,854; integer-verified).

Cross-tabulating the disease axis against provenance reveals a structural confound (Table 1, from the dataset's own metadata and confirmed against the paper's Methods):

Stage	n donors	Tissue source	Lobes
Healthy	4	surgical / atlas (mixed; one with biliary-stone disease)	mixed
NAFLD	7	needle biopsy	right
NASH (no cirrhosis)	27	needle biopsy	right
NASH cirrhosis	4	needle biopsy	right
End-stage	5	transplant explant	left + right + caudate

Every end-stage sample is an explant; every milder disease sample is a right-lobe needle biopsy. The paper's own Methods confirm this (16-gauge needle biopsies for disease; "healthy donors and explant tissue" sampled as ~1 cm³ cubes from three lobes for end-stage), and report no ischemia times.

Procurement and ischemia induce an acute stress program — and this is not an incidental observation but the *documented reason* single-nucleus data from explanted organs cannot be read alongside biopsies. Tissue dissociation and warm ischemia are long-established inducers of an immediate-early (*FOS/JUN*-family) and heat-shock artifact that is highly conserved across cell types, distorts the measured transcriptome, and was itself the cause of the original spurious single-cell "stress subpopulations"; it persists in single-nucleus protocols [1–3]. Summing raw UMIs of immediate-early (*FOS, JUN, JUNB, JUND, ATF3, DUSP1*) and heat-shock (*HSPA1A, HSPA1B*) genes per nucleus, this signature is elevated **5–21× in every lineage** in explants versus biopsies — hepatocytes 21×, **endothelial cells 20×**, stellate 13×, macrophage 9×, cholangiocyte 9×, lymphocyte 4× (Fig 1). That endothelial cells, irrelevant to hepatocyte zonation, are as stressed as hepatocytes establishes the signal as **organ-level handling/ischemia, not a hepatocyte disease program**. End-stage depth is not low (median raw library ~5,500), so this is active stress transcription, not dropout.

Critically, this artifact threatens **the zonation readout specifically**, not just overall data quality. Hepatocyte metabolic zonation is maintained by a *standing* oxygen gradient and a Wnt/R-spondin morphogen gradient supplied by central-vein endothelium, and the oxygen-dependent pericentral detox program — the *CYP/ADH* axis on which the de-zonation claim rests — is the most perfusion-sensitive part of it [4]. Whole-organ explantation collapses perfusion and oxygenation and ischemically shocks that endothelial source, so the very pericentral program we score is precisely the one procurement is expected to perturb: **an apparent end-stage de-zonation is exactly the artifact this confound would manufacture**, independent of any disease. (A hypoxia/HIF panel — *VEGFA, SLC2A1, LDHA, CA9, ENO1, PGK1* — is, by contrast, only modestly and non-uniformly elevated, hepatocytes 1.2×, arguing against a *sustained canonical HIF program* in the nuclei but not

excluding acute ischemic injury.) Even the least-stressed explant hepatocytes carry far less detox signal than biopsy F4 (≈ 2 vs ≈ 7 molecules per nucleus), so the explant state cannot be subtracted out. The exclusion of end-stage therefore rests on three independent legs: the perfect stage $\perp$ tissue-source collinearity (Table 1), the organ-wide cross-lineage stress (above), and the prior literature establishing that this exact stress artifact both distorts snRNA expression [1–3] and falls hardest on the perfusion-dependent zonation axis we measure [4].

We therefore exclude end-stage as a disease endpoint and restrict the disease analysis to the acquisition-matched needle-biopsy donors (fibrosis Fo–F4; NAS activity), keeping explants only as a heterogeneous contrast (§3), not a clean positive control.

2. No detectable de-zonation across biopsy MASLD, by any count-based mechanism

We enumerated every marker-count mechanism by which zonation could collapse that is *detectable in spatially-blind single-nucleus data* (true spatial relocation, focal peri-fibrotic remodeling, and lobular architecture are not testable here — see Limitations) and mapped each to an absolute, count-based signature (Table 2). All analyses use raw counts, **depth-matched** by downsampling each nucleus to a common UMI budget (Monte-Carlo binomial thinning; standard in single-cell practice), with the **donor** as the unit of inference.

Scenario	Count signature	Biopsy Fo–F4	NAS
Pericentral-cell depletion	PC-anchor nuclei ↓	no change	no change
Within-cell dimming	detox within PC nuclei ↓	no robust change (§4)	no robust change
De-zonation (co-expression)	dual PC ⁺ PP ⁺ nuclei ↑	no change	no change
De-zonation (gradient compression)	per-nucleus PC/(PC+PP) → middle	no change	no change
Identity turn-off	double-negative nuclei ↑	no change	no change
Compositional shift	PP:PC anchor ratio	no change	no change

We classified each hepatocyte nucleus as pericentral-anchor (*GLUL* and/or *CYP3A4*), periportal-anchor (≥ 2 of *ASS1/PCK1/HAL/ALDOB*), dual, or null — the canonical zonation landmarks, including the *GLUL/ASS1* axis on which the original de-zonation figure rests. Across biopsy Fo–F4 the pericentral-anchor fraction is **flat and non-monotone** (F1 is the floor, not F4), and this holds under: anchor = *GLUL*-only / *CYP3A4*-only / both; depth budget $B = 1,000 / 1,500 / 3,000$; and multiple thresholds. Averaging over eight thinning draws, the Monte-Carlo noise on the donor-median is SD 0.006–0.010 — negligible against the effect we could detect.

De-zonation as co-expression: the dual (PC⁺PP⁺) fraction shows no fibrosis trend at any depth budget — the F1→F4 slope is near-zero and non-monotone across $B = 1,000/1,500/3,000$ (its absolute *level* scales with depth, an ambient-detection effect; its disease *trend* does not). At a stringent ≥ 2 -UMI threshold, biopsy dual is ~ 0.003 – 0.006 at every fibrosis stage, whereas the explant group is **0.029 — roughly 7× higher** and surviving the ambient-robust cut. So genuine hepatocyte co-expression is near-absent across biopsy disease and an order of magnitude higher

only in explants. **De-zonation as gradient compression** (the classical reading — cells losing their zonal extremes and drifting to the middle), measured as the per-nucleus zonal balance PC/(PC+PP) in absolute count-bins above an informative floor (sensitivity $q = 2/3/5$), shows **stable poles and a stable midpoint** (informative denominator 0.98–1.00 in every donor); the only movement is a mild, non-monotone redistribution from PC-leaning toward PP-leaning that peaks at F3 and reverts at F4. **Turn-off** (double-negative nuclei) and **compositional shift** (PP:PC ratio ~1.0) are likewise flat. The two-dimensional joint distribution of pericentral- versus periportal-program counts has the same shape across Fo–F4 and collapses to a single periportal pole only in explants (Fig 2).

Why canonical anchors — and is the null marker-set-dependent? We lead with a small canonical landmark set for one reason: every gene is an auditable, literature-assigned zonation marker and the census is a direct molecule count, which is exactly what lets us avoid Paper 2's *fitted relative* pseudo-coordinate — the z-scored signature ruler that converts depth, cell number and tissue state into an apparent collapse, the instrument we reject (Provenance). The choice of set is therefore principled, but it must not *drive* the result. So we re-ran the identical depth-matched, donor-level count census on three sets of increasing breadth: the 2/4-gene canonical anchors; a 13-gene curated program; and **Paper 2's own 20 + 20 hepatocyte landmark genes** (extracted verbatim from their spatial-atlas ruler and used as gene *identities* in a detection census, never as their η coordinate). On set-size-normalized endpoints the Fo→F4 biopsy profile is **flat and non-monotone in all three** (`markerset_robustness.csv`): per-nucleus pericentral share across Fo–F4 is 0.36/0.19/0.27/0.24/0.26 (canonical), 0.66/0.65/0.61/0.58/0.59 (program), and **0.56/0.54/0.51/0.47/0.49 (Paper 2's landmarks)** — no progressive loss of polarity on any set, including Paper 2's. (Their landmark set does not even contain *GLUL* — they too treat it as snRNA dropout-prone — so the null does not hinge on any single fragile marker.) The negative is thus a property of the data, not of our marker choice.

Power is the load-bearing caveat, and we put a number on it. The MDE below is **a calibration of statistical power, not a biological result** — it does not measure zonation, it measures what this cohort *could have seen*. We use it for one purpose only: to **bound the negative** — to say which large effects a biopsy of this size would have been able to detect. We never read it as evidence that zonation is preserved; a small MDE would still not license that claim. With the **donor** as the unit and small end strata (Fo $n=2$, F4 $n=4$, each ~59% one donor), the two-sample minimum detectable effect is

$$\text{MDE} = (z_{1-\alpha/2} + z_{\text{power}}) \cdot s \cdot \sqrt{(1/n_1 + 1/n_2)} = 2.80 \cdot s \cdot \sqrt{(1/n_1 + 1/n_2)} \quad (\alpha=0.05 \text{ two-sided, } 80\% \text{ power; } s = \text{pooled donor-level SD}),$$

reported **separately for every count endpoint** — one MDE per candidate de-zonation mechanism — from the actual donor-level SDs across the F1→F4 disease span (`mde_table.csv`):

Mechanism (de-zonation as...)	Endpoint	Mean	Donor SD	n (F1/F4)	MDE (abs)	What this rules out
pericentral depletion	PC-anchor fraction	0.27	0.13	8/4	0.22	only a near-total pericentral collapse ($\approx 80\%$ relative loss); not a moderate decline
periportal depletion	PP-anchor fraction	0.23	0.10	8/4	0.17	only a large ($\sim 70\%$) periportal-anchor shift
turn-off (double-negative)	null fraction	0.40	0.17	8/4	0.29	only a large rise in marker-negative nuclei
co-expression	dual (≥ 2 UMI) fraction	0.006	0.010	8/4	0.017	only a <i>large</i> dual rise (the dramatic claim); subtle co-expression not excluded
gradient compression	zonal-extreme fraction	0.47	0.072	8/4	0.12	only a ≈ 0.12 ($\sim 26\%$) drift off the zonal poles; finer compression not excluded
within-cell dimming	within-PC detox burden (UMIs/nuc)	10.7	1.6	8/4	2.78	a $\approx 26\%$ within-PC change is detectable — but the targeted dip it would catch is not confirmed by genome-wide DE (§4), so we do not claim it

Read row by row: the **structural** endpoints (PC/PP depletion, turn-off) are powered only to exclude large-to-near-total shifts (MDE $\approx 70\text{--}110\%$ of the mean), so the negative there rules out *collapse*, not subtle change. **Co-expression** is a near-zero endpoint, so the informative quantity is the *absolute* MDE ≈ 0.017 — we exclude the dramatic dual-positive rise but not a small one. **Gradient compression** and **within-cell dimming** are the better-powered endpoints (MDE $\approx 26\%$): the within-PC dimming endpoint is sensitive enough to have caught the $\sim 30\%$ targeted dip, yet that dip does *not* survive genome-wide donor-level testing (§4) — so here the better power cuts the other way, telling us the absence of a robust detox change is not a power artifact. The interior F1 $\leftrightarrow$ F3 contrast ($n = 8/12$) tightens each of these by $\sim 40\%$ (e.g. PC-anchor MDE $0.22 \rightarrow 0.11$, gradient $0.12 \rightarrow 0.10$) and is where the negative is best supported. In every case the MDE **excludes near-total marker-count collapse but cannot quantify preservation, and does not rule out moderate, focal, or spatially restricted de-zonation** — it is reported here as a limitation and bounding device, cross-referenced in Limitations, never as a positive claim.

The load-bearing evidence is the donor-level absolute-count table (depth-matched; per-donor numerators and denominators, no pooling, no fold-changes; full per-donor table in `load_bearing_donor_table.csv`):

group	donors	median nuclei/donor	PC-anchor	dual (≥ 2 UMI)	null	stress / 10^4
F0 biopsy	2	935	36%	0.05%	34%	0.3
F1 biopsy	8	878	19%	0.23%	44%	0.4
F2 biopsy	12	721	23%	0.45%	36%	0.5
F3 biopsy	12	850	22%	0.21%	39%	0.4
F4 biopsy	4	472	21%	0.24%	39%	0.5
explant CL103	1	1,960	14%	2.7%	26%	7.9
explant CL104	1	1,908	50%	1.1%	18%	11.9
explant CL16	1	5,955	3%	2.4%	18%	20.2
explant CL17	1	4,111	9%	5.9%	17%	6.1
explant CL18	1	8,285	20%	22.4%	5%	10.2

Across biopsy fibrosis the per-donor counts are flat and the stress burden is uniformly low (~ 0.3 – $0.5/10^4$); the five explants are each abnormal but in *different* directions (one retains 50% pericentral anchoring, one collapses to 3%, one explodes to 22% co-expression) and uniformly high-stress (6–20/ 10^4). The flat biopsy census and the heterogeneous, stress-high explants are both visible here without any relative statistic.

Confound controls, in counts (diagnostics, not full structural kills). Four count-based controls are consistent with the biopsy census being real, not technical. (i) *Donor-specific ambient RNA*: per-donor soup scales with the most abundant transcript (*ALB*, 17–102 per 10^4 across biopsy donors), but the dual fraction does not track it (rank correlation ≈ 0 over 38 donors) — a diagnostic, not a full ambient correction (SoupX/empty-droplet profiles were unavailable). (ii) *Cholangiocyte mis-annotation*: across the **full biliary/plasticity panel** (*KRT7*, *KRT19*, *KRT23*, *EPCAM*, *SOX4*, *NCAM1*, *MUC1*, *BCL2*), dual and periportal-pole nuclei are not enriched (all ≤ 0.004) relative to pericentral nuclei, so they are not contaminating cholangiocytes. (iii) *Depth/complexity*: median features per nucleus — a depth/complexity proxy, **not true ploidy** (which is unmeasured here) — is non-monotone with fibrosis (2,218 / 3,063 / 3,169 / 2,641 / 1,984 for F0–F4) rather than tracking stage. (iv) *Depth-matching discard*: the 3.5–7.8% of nuclei dropped below budget are uniformly lower-detection at every stage, so the discard is depth-driven and does not bias the F0→F4 comparison. None manufactures the null; none is a complete structural kill.

3. The "collapse" is confined to explants — and is heterogeneous procurement damage, not a uniform disease state

The metrics that are flat across biopsy disease all move in the explant group — but **not concordantly**. The five explants are not one phenotype: per-donor pericentral-anchor fraction ranges 0.03–0.50 and periportal-to-pericentral ratio ranges 0.15–20. One explant (CL16) shows an extreme periportal-pole collapse (PP:PC ≈ 20); another (CL104) *retains* more pericentral anchoring

(0.50) than the average biopsy donor (0.24); a third (CL18) shows a co-expression explosion (dual fraction 0.56). This is the signature of **heterogeneous procurement/ischemic damage across five separately-handled organs**, not a coherent end-stage disease program — and it is exactly the pooled-summary-hiding-heterogeneity that the original marker-correlation collapses into a single number.

A within-explant stress association is only suggestive: per-donor (not pooled — to avoid pseudoreplication), the pericentral-anchor difference between each donor's low- and high-stress halves is $-0.03, -0.05, +0.00, +0.03, -0.03$ — i.e. 3 of 5 donors in the predicted (collapse-with-stress) direction, two against. So procurement damage is multi-factorial (ischemia time, handling, organ state) rather than a single clean stress-dose axis; we report the association as suggestive, not as an identified mechanism. Either way, because disease stage is perfectly collinear with explant source and the damage is organ-wide (§1), the explant signal cannot be attributed to disease, and we treat it only as a contrast — not a disease endpoint and not a clean positive control.

4. Genome-wide, the biopsy hepatocyte transcriptome is broadly stable — the only intrinsic signal is a modest acute-phase response, not de-zonation

The marker-count census above is targeted by design. To ask whether *anything* else changes — and in particular whether the zonation/xenobiotic programs move when tested without a curated panel — we ran **transcriptome-wide donor-level differential expression**: pseudobulk of raw UMIs over each donor's hepatocyte nuclei (38 biopsy donors; 17,572 genes expressed in $\geq$ half the donors), Spearman trend against fibrosis stage with Benjamini–Hochberg control (`dge_genomewide.csv`). Two results.

First, **the zonation and xenobiotic programs are flat genome-wide**. Every pericentral/periportal and detox gene is non-significant — *CYP2E1* FDR 0.98, *CYP1A2* 0.91, *ADH4* 0.97, *SLCO1B3* 0.80, *CYP3A4* 0.97, *GLUL* 0.85, and all periportal genes FDR > 0.79 , with $|\log_2FC| \leq 0.4$. The de-zonation null therefore holds not just on our anchors but across the whole transcriptome. It also **supersedes a targeted observation we had over-weighted**: within pericentral-anchor nuclei the depth-matched xenobiotic burden declines modestly (12.7 $\rightarrow$ 8.9 molecules/nucleus F1 $\rightarrow$ F4), but that within-subpopulation dip **does not survive the genome-wide donor-level test** (the same genes sit at FDR ≈ 0.8 –0.98). We therefore no longer single it out as a finding; at most it is a weak, conditioned, exposure-confounded signal (these genes also respond to alcohol/*CYP2E1*, smoking/*CYP1A2*, bile-acid–FXR/*SLCO1B3*, *AKR1D1*, none recorded here), and it is not a zonation result.

Second, **what does trend with fibrosis is not a hepatocyte zonation program**. Twenty-three genes reach FDR < 0.05 . A compositional audit (`dge_compositional.csv` : per-lineage burden across all 99,809 nuclei, plus per-donor tracking of the stromal fraction) shows **15 of the 23 are expressed $\geq 3\times$ more highly in a non-hepatocyte lineage** (cholangiocyte, stellate, endothelial or immune) than in hepatocytes — e.g. *ADAMTS9* and *GUCY1A1* (stellate ≈ 15 –17 $\times$), *EPB41L4A* (endothelial $\approx 7\times$), several cholangiocyte-dominated — and are most consistent with low-level **ambient contamination** of hepatocyte nuclei (the non-hepatocyte fraction itself does *not* rise with fibrosis here, $\rho = -0.07$, so this is per-nucleus ambient, not a gross composition shift). The genuinely hepatocyte-intrinsic trend is led by **A2M** (acute-phase $\alpha 2$ -macroglobulin; hepatocyte-dominant; CPM 219 $\rightarrow$ 718), with a few others — i.e. a **modest hepatocyte acute-phase/injury response** that rises with fibrosis. Progenitor/biliary markers (*SOX9*, *KRT19*, *SOX4*) stay at

background (~ 0.001 – 0.005) across all stages, arguing against frank dedifferentiation. The net biopsy-internal picture is thus a broadly stable hepatocyte transcriptome with a small acute-phase signal — **no robust zonation or xenobiotic change of any kind**.

Discussion

Two claims, kept deliberately separate:

A — the answer to the question. Across acquisition-matched biopsy MASLD, spanning Fo to cirrhotic F4 and the orthogonal NAS activity axis, there is **no detectable hepatocyte de-zonation by any count-based mechanism**, robust to depth budget, draw, anchor definition and thresholds. The study is powered to exclude near-total loss, not finer change. This null is also what biology predicts: hepatocyte zonation is maintained by a *standing* oxygen gradient and a Wnt/Rspo morphogen gradient supplied by central-vein endothelium [4] (canonically with *GLUL* as the β -catenin master output — invoked here as established biology of the gradient, not as our evidence, since *GLUL* is dropout-prone in snRNA and we do not rely on its detection). A needle biopsy of pre-cirrhotic-to-nodular parenchyma still contains central veins and that gradient source, so surviving hepatocytes should remain zoned — whereas whole-organ explantation ischemically shocks the source, which is precisely the organ-wide, donor-specific, stress-co-varying collapse we see only in the explants (§1, §3). The negative is therefore biologically *plausible* — it is the count signature expected if surviving hepatocyte pericentral/periportal programs remain separable in sampled parenchyma — and the explant collapse is *consistent with* procurement stress. We say plausible and consistent-with, not proven: a needle core of cirrhotic tissue may or may not capture central veins cleanly, and fibrotic septa, nodules, altered perfusion and local endothelial remodeling could disrupt spatial programs without producing a clean global co-expression signature. The data show the end-stage signal is **inseparable from explant/procurement stress** — not that it is non-disease.

B — what we do not claim. The original de-zonation conclusion rests on three evidence legs — snRNA marker-correlation, immunofluorescence (*GLUL*+*ASS1* co-staining), and 3D whole-organ FLASH imaging — **all on end-stage explant tissue**. Our re-analysis corrects only the first: the transcriptional signal is a relative-correlation statistic measured on procurement-confounded tissue. We do not, and cannot, refute the protein-imaging legs: protein co-localization operates on a different molecule and timescale than the acute transcriptional stress we document, and end-stage cirrhosis genuinely remodels lobular architecture. What our data *do* contribute to that discussion is a **staging argument**: if de-zonation were a progressive disease process, cells would drift toward co-expression by cirrhotic F4 biopsy, and none do (dual ≥ 2 -UMI ≈ 0.003 – 0.005 at every stage, non-monotone — indistinguishable from zero). The end-stage co-expression is thus either genuine late-cirrhotic remodeling or explant confound — undecidable here — but it is **not reached by a detectable progressive trajectory**.

C — the transdifferentiation link. The original study's headline is hepatocyte $\leftrightarrow$ cholangiocyte plasticity, and a natural question is whether de-zonation accompanies it. Within hepatocyte nuclei, biliary/transdifferentiation markers (*KRT7*, *KRT19*, *EPCAM*, *SOX4*) stay at background (~ 0.4 – 1.3% detection) and flat across biopsy F1–F4, and rise only in end-stage explants — co-located with the apparent de-zonation. So in acquisition-matched tissue neither phenomenon is detectable, and any co-occurrence at end-stage is inseparable from the shared procurement confound. We therefore cannot establish (nor refute) a zonation-degradation $\leftrightarrow$ transdifferentiation link; we note only that

the hepatocyte-side biliary signature, like the de-zonation signal, is confined to the confounded explant group. This does not address the study's organoid/cholangiocyte-side evidence for plasticity.

The broader methodological point is that **relative summary statistics (marker correlations, spreads, z-scores) on cohorts where disease stage tracks tissue acquisition can manufacture biology**. Count-based, depth-matched, donor-level analysis with an explicit confound audit and an enumerated scenario taxonomy is a more conservative standard.

Limitations

- **Power:** a bounded null. The per-endpoint MDE table (§2) is a calibration of statistical power, not a biological result; it bounds the negative — near-total marker-count loss is excluded, but moderate, focal, or graded change is not, and a small MDE is never read as preservation.
- **Tissue source:** end-stage disease is unrecoverable here (no end-stage biopsies, no non-end-stage explants).
- **Modality:** single-nucleus transcript counts cannot see lobular spatial position; "PC-like program," not location. Focal peri-fibrotic remodeling, true spatial relocation, pericentral-substate replacement, and functional/enzymatic zonation are not testable here.
- **Markers:** *GLUL* is dropout-prone in snRNA and *CYP3A4* covaries with the detox program; we therefore make no independent "identity-preserved" claim — only the structural null.
- **Metric weighting:** the gradient/polarization metric is an *unweighted* sum of curated panel genes, so high-expressers (e.g. *ALDOB*) dominate the pericentral ↔ periportal balance. We therefore treat the polarization as a coarse structural check (does the joint distribution shift or compress?) rather than a calibrated coordinate, and cross-check every conclusion against the threshold-based anchor census, which does not pool across genes.
- **Covariates:** the recorded demographics (age, sex, type-2 diabetes) were tested and do not explain the weak §4 within-PC xenobiotic dip (which in any case does not survive genome-wide testing); but etiology, alcohol, smoking, medication and fasting are *not* recorded and cannot be ruled out — so no disease-specific interpretation of those transcripts is warranted (§4).
- **Replication:** single cohort; no independent biopsy series or protein validation.

Methods

Dataset. GSE202379 (Gribben et al., Nature 2024): single-nucleus RNA-seq of 47 human liver donors across MASLD stages (healthy $n=4$, NAFLD 7, NASH-no-cirrhosis 27, NASH-cirrhosis 4, end-stage 5), 69,426 nuclei annotated as hepatocytes plus 30,383 non-hepatocyte nuclei (endothelial, stellate, cholangiocyte, macrophage, lymphocyte). Disease staging (fibrosis Fo–F4; steatosis/ballooning/inflammation → NAS) and provenance fields (donor ID, lobe) are in the per-cell metadata.

Raw-count extraction. The deposited Seurat object contains both a raw **RNA** assay and an **SCT** assay (SCTransform-normalized), with **SCT** set as the *default* assay. SCTransform is a legitimate normalization for the authors' downstream tasks (clustering, dimensional reduction, differential testing), but its **counts** slot holds depth-equalized *corrected* counts — a model output, not observed molecules: per-nucleus SCT totals equal **nCount_SCT** exactly (Pearson $r = 1.000$) and are compressed to ~3,000–5,700, whereas raw **nCount_RNA** spans 935–49,854. Molecule-level inference (detection

fractions, co-expression, our own depth control) therefore requires the **raw RNA assay**, not the corrected default. We loaded the original object (`GSE202379_SeuratObject_AllCells.rds`) in R (SeuratObject 5.4) and extracted the raw `counts` layer of the RNA assay for a curated ~45-gene panel (pericentral identity *GLUL*, *CYP3A4*; pericentral xenobiotic *CYP2E1*, *CYP1A2*, *ADH1B*, *ADH4*, *AKR1D1*, *SLCO1B3*; Wnt-target *LGR5*, *AXIN2*; periportal *ASS1*, *CPS1*, *PCK1*, *ALDOB*, *HAL*, *ARG1*; immediate-early *FOS*, *JUN*, *JUNB*, *JUND*, *ATF3*, *DUSP1*; heat-shock *HSPA1A*, *HSPA1B*, *HSP90AA1*; HIF/hypoxia *VEGFA*, *SLC2A1*, *LDHA*, *CA9*, *ENO1*, *PGK1*; housekeeping/identity *ACTB*, *GAPDH*, *B2M*, *ALB*, *TTR*, *APOA1*; biliary/plasticity *KRT7*, *KRT19*, *EPCAM*, *SOX4*, *KRT23*, *NCAM1*, *MUC1*, *BCL2*), together with each nucleus's full raw library size (`E_raw`). Counts were integer-verified; library sizes 920–49,854.

Provenance / confound audit. Disease stage was cross-tabulated against tissue source and lobe from the metadata and confirmed against the paper's Methods. Donor-ID prefixes and lobe sampling identify all end-stage donors as multi-lobe transplant explants and all milder-disease donors as right-lobe needle biopsies; healthy donors are a mix of surgical/atlas tissue (one with biliary-stone disease). All disease donors share a single experiment label, so technical batch is perfectly confounded with donor and tissue source and cannot be separated here — untestable, not absent.

Stress and hypoxia scores. For each nucleus we summed raw UMIs of the immediate-early + heat-shock panel ("procurement stress") and, separately, of the HIF-target panel ("hypoxia"). Per donor and cell lineage we computed UMIs per 10^4 of that lineage's total, then took the **median across donors** (the explant/biopsy ratios are reported on this donor-median estimator; a pooled-sum estimator gives smaller ratios because two high-depth explants dominate the pool).

Depth-matching (the core control). Detection and co-expression rise with sequencing depth, which differs by donor and (slightly) by stage. We removed this by downsampling every nucleus to a common budget B via binomial thinning — each panel-gene count retained with probability B/E_raw , using the nucleus's *full* RNA library size as the sampling denominator (a marginal approximation to full-transcriptome multinomial downsampling, exact for the per-gene detection of the rare markers that drive the census) — discarding nuclei below B (~4–8%, highest in the shallower F4 stratum; the discard is depth-driven and consistent across stages). All per-nucleus quantities were averaged over 8 independent thinning draws (Monte-Carlo SD on donor-median fractions = 0.006–0.010). Primary $B = 1,500$; results reported across $B \in \{1,000, 1,500, 3,000\}$. This is the standard count-preserving depth normalization in single-cell analysis (e.g. `DropletUtils::downsampleMatrix`, `10X cellranger aggr --normalize=mapped`).

Anchor census. On depth-matched counts, each hepatocyte nucleus was classified: **pericentral-anchor** = detection of *GLUL* and/or *CYP3A4*; **periportal-anchor** = ≥ 2 of $\{ASS1, PCK1, HAL, ALDOB\}$; **dual** = both; **null** = neither. These are the canonical zonation landmarks, including the *GLUL/ASS1* axis on which the original de-zonation figure rests. Robustness was assessed over: anchor definition (*GLUL*-only / *CYP3A4*-only / both), per-gene UMI threshold $k \in \{1, 2\}$, periportal rule (2-of-4 / 3-of-4, *ALDOB* in/out, *CPS1*-based), and depth budget. *GLUL* is dropout-prone in snRNA (detection ~0.18) and *CYP3A4* covaries with the xenobiotic genes (Pearson +0.53), so no independent "identity-preserved" claim is made — only the structural census. **Marker-set robustness:** the census was repeated on a 13-gene curated program and on Paper 2's exact 20 + 20 hepatocyte landmark genes (Yakubovsky et al. [5]; extracted from the original Seurat RNA assay by `prep/08_extract_paper2_landmark_raw.R`, all 40 present in Paper 1), scored on set-size-normalized endpoints (per-nucleus pericentral share and zonal-extreme fraction); the Fo–F4 null held on every

set (`markerset_robustness.csv`). We use Paper 2's landmark gene *identities* in this count census but never their fitted η pseudo-coordinate.

Gradient (compression) test. Per nucleus we summed a pericentral program and a periportal program (depth-matched) and, among informative nuclei (program sum $\geq q$; $q \in \{2,3,5\}$), computed the zonal balance PC/(PC+PP) and binned it into absolute count classes (PC-pole / PC-leaning / middle / PP-leaning / PP-pole). Compression (de-zonation) would shrink the poles and grow the middle while the informative denominator held; we report per-stage class fractions and the informative-denominator range across donors.

Within-pericentral attenuation. Within nuclei classified pericentral-anchor by *GLUL/CYP3A4* (excluding the genes under test), we summed depth-matched xenobiotic counts per nucleus and summarized per donor (median and range) by stage, to separate per-cell dimming from pericentral-cell depletion. This targeted, conditioned estimator is reported only in the context of the genome-wide test below, which does not confirm it.

Genome-wide differential expression (the "is anything else changing?" test). From the original Seurat object we extracted full-transcriptome raw RNA counts and formed a **donor pseudobulk** — the sum of raw UMIs over each donor's hepatocyte nuclei (`prep/09` ; 31,257 genes $\times$ 47 donors). Restricting to acquisition-matched biopsy donors ($n = 38$; Fo–F4 = 2/8/12/12/4) and to genes expressed (CPM ≥ 1) in at least half of them (17,572 genes), we tested each gene for a fibrosis trend by Spearman correlation of log₂-CPM against stage across donors, with Benjamini–Hochberg FDR, plus an F1-vs-F4 Mann–Whitney contrast (`dge_genomewide.py` $\rightarrow$ `dge_genomewide.csv`). The unit is the donor; this is the count-based, donor-level analogue of a pseudobulk DE and avoids per-nucleus pseudoreplication.

Compositional audit of the DE hits. For every FDR < 0.05 gene we asked whether the signal is hepatocyte-intrinsic or ambient: we extracted that gene's raw counts across **all 99,809 nuclei** (`prep/10`) and computed its depth-normalized burden (UMIs/ 10^4) per cell lineage, flagging genes $\geq 3\times$ more expressed in a non-hepatocyte lineage; and we tested, across biopsy donors, whether the gene's hepatocyte-pseudobulk CPM tracked the donor's non-hepatocyte nuclei fraction (`dge_compositional.py` $\rightarrow$ `dge_compositional.csv`).

Two-dimensional joint distribution. Per stage, the depth-matched pericentral-program vs periportal-program counts were rendered as a donor-balanced 2-D histogram (≤ 300 nuclei/donor) to visualize whether the joint distribution shifts, compresses, or collapses.

Disease axes and inference. The **donor** is the unit of inference throughout; per-donor values were summarized as medians and ranges across donors within a stratum. Two orthogonal disease axes were used — fibrosis Fo–F4 and NAS activity (steatosis + ballooning + inflammation) — both restricted to needle-biopsy donors; end-stage explants and the mixed-source healthy group were excluded from the disease axis and retained only as contrasts. Minimum detectable effects were estimated at the donor level (two-sample, $\alpha=0.05$, 80% power) from the within-stratum donor standard deviation.

Code and figures. Raw extraction `src/05_extract_raw_panel.R` (`08_extract_paper2_landmark_raw.R` for Paper 2's landmark genes; `09_extract_full_and_union.R` for the full-transcriptome pseudobulk + curated-set per-cell raw; `10_extract_dge_genes_allcells.R` for the DE-hit cross-lineage counts); analysis `src/analysis/{raw_counts, census, census_2d, gapfill, census_v2, review_checks, supplementary_checks, load_bearing, mde, markerset_robustness,`

`all_sets_census`, `dge_genomewide`, `dge_compositional`}.py (`load_bearing` / `mde` produce the donor-level absolute-count table and per-endpoint MDE table; `markerset_robustness` / `all_sets_census` the set-invariance checks; `dge_genomewide` / `dge_compositional` the transcriptome-wide test and its compositional audit); sanity/provenance `src/prep/{06_sanity_raw.py, 07_dump_object_schema.R}` (the latter cover the within-explant dose-response, per-donor explant heterogeneity, ambient/contamination/ploidy confounder tests, the cross-lineage stress/HIF ratios, and the recorded-covariate test); figures `results/figures/h2/{gapfill_polarization, census_2d_joint}.png`. Full numeric record in `FINDINGS.md` and the per-analysis CSVs under `results/tables/analysis/` (incl. `load_bearing_donor_table.csv`, `mde_table.csv`, `markerset_robustness.csv`).

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